

ALFAAS37188

## Paraformaldehyde, 97%

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                                      |                                                                                                                                                                                                                                                                    |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description:                        | Paraformaldehyde, 97%<br>Paraformaldehyde, 97%                                                                                                                                                                                                                     |
| Cat No. :<br>Synonyms<br>CAS No<br>Molecular Formula | S37188<br>Formaldehyde polymer; Polyoxymethylene; Polyformaldehyde<br>30525-89-4<br>(CH <sub>2</sub> O) <sub>n</sub>                                                                                                                                               |
| Supplier                                             | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| Emergency Telephone Number                           | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                                       | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use<br>Uses advised against              | Laboratory chemicals.<br>No Information available                                                                                                                                                                                                                  |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                                                                                                                                                                                                                                             |                     |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------|
| Physical State<br>Solid                                                                                                                                                                                                                                                                                                                                                     | Appearance<br>White | Odor<br>pungent |
| <b>Emergency Overview</b><br>Flammable solid. Causes skin irritation. Causes damage to organs. May cause respiratory irritation. Harmful to aquatic life with long lasting effects. Harmful if swallowed. May cause an allergic skin reaction. Causes serious eye damage. Harmful if inhaled. Suspected of causing cancer. May form combustible dust concentrations in air. |                     |                 |

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Flammable solids.                                  | Category 2 |
| Acute Oral Toxicity                                | Category 4 |
| Acute Inhalation Toxicity - Dusts and Mists        | Category 4 |
| Skin Corrosion/Irritation                          | Category 2 |
| Serious Eye Damage/Eye Irritation                  | Category 1 |
| Skin Sensitization                                 | Category 1 |
| Carcinogenicity                                    | Category 2 |
| Specific target organ toxicity - (single exposure) | Category 1 |
|                                                    | Category 3 |
| Chronic aquatic toxicity                           | Category 3 |

#### Label Elements

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## Signal Word

## Danger

## Hazard Statements

H228 - Flammable solid  
H302 + H332 - Harmful if swallowed or if inhaled  
H315 - Causes skin irritation  
H317 - May cause an allergic skin reaction  
H318 - Causes serious eye damage  
H335 - May cause respiratory irritation  
H351 - Suspected of causing cancer  
H370 - Causes damage to organs  
H412 - Harmful to aquatic life with long lasting effects

## Precautionary Statements

## Prevention

P201 - Obtain special instructions before use  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P271 - Use only outdoors or in a well-ventilated area  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves

## Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

Combustible material. May form combustible dust concentrations in air.

## Health Hazards

Causes skin irritation. Causes damage to organs. May cause respiratory irritation. Harmful if swallowed. May cause an allergic skin reaction. Harmful if inhaled. Suspected of causing cancer.

## Environmental hazards

Harmful to aquatic life with long lasting effects. Is not likely mobile in the environment due its low water solubility. .

May form explosible dust-air mixture if dispersed. Toxic to terrestrial vertebrates. Toxicity to Soil Dwelling Organisms. This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component        | CAS No     | Weight % |
|------------------|------------|----------|
| Paraformaldehyde | 30525-89-4 | >90      |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

Causes severe eye damage. May cause allergic skin reaction. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Self-Protection of the First Aider**

Use personal protective equipment as required.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Flammable. Containers may explode when heated. Fine dust dispersed in air may ignite. Combustible material.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Remove all sources of ignition.

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7. HANDLING AND STORAGE

### Handling

Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Ensure adequate ventilation. Keep away from open flames, hot surfaces and sources of ignition.

### Storage

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

### Specific Use(s)

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

### Exposure Controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Nitrile rubber |                                   |                 |             |                       |
| Neoprene       |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Skin and body protection

Long sleeved clothing

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits

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are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Particle filtering: EN149:2001

When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

No information available.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                                |                                             |                                          |
|------------------------------------------------|---------------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | White                                       |                                          |
| <b>Physical State</b>                          | Solid                                       |                                          |
| <b>Odor</b>                                    | pungent                                     |                                          |
| <b>Odor Threshold</b>                          | No data available                           |                                          |
| <b>pH</b>                                      | 3.5-5.0                                     | 10% susp                                 |
| <b>Melting Point/Range</b>                     | 120 - 170 °C / 248 - 338 °F                 |                                          |
| <b>Softening Point</b>                         | No data available                           |                                          |
| <b>Boiling Point/Range</b>                     | No information available                    |                                          |
| <b>Flash Point</b>                             | 71 °C / 159.8 °F                            | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable                              | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available                    |                                          |
| <b>Explosion Limits</b>                        | <b>Lower</b> 7 vol%<br><b>Upper</b> 73 vol% |                                          |
| <b>Vapor Pressure</b>                          | 1.2 mmHg @ 25 °C                            |                                          |
| <b>Vapor Density</b>                           | Not applicable                              | Solid                                    |
| <b>Specific Gravity / Density</b>              | 1.46                                        |                                          |
| <b>Bulk Density</b>                            | No data available                           |                                          |
| <b>Water Solubility</b>                        | Slightly soluble                            |                                          |
| <b>Solubility in other solvents</b>            | No information available                    |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                             |                                          |
| <b>Autoignition Temperature</b>                | 300 - °C / 572 - °F                         |                                          |
| <b>Decomposition Temperature</b>               | No data available                           |                                          |
| <b>Viscosity</b>                               | Not applicable                              | Solid                                    |
| <b>Explosive Properties</b>                    |                                             | explosive air/vapour mixtures possible   |
| <b>Oxidizing Properties</b>                    | No information available                    |                                          |
| <b>Molecular Formula</b>                       | (CH <sub>2</sub> O) <sub>n</sub>            |                                          |

**SECTION 10. STABILITY AND REACTIVITY**

|                                         |                                                                                                      |
|-----------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Stable under normal conditions.                                                                      |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                        |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                             |
| <b>Conditions to Avoid</b>              | Excess heat. Avoid dust formation. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Materials to avoid</b>               | Strong oxidizing agents.                                                                             |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                             |

**SECTION 11. TOXICOLOGICAL INFORMATION**

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## Product Information

## (a) acute toxicity;

| Component        | LD50 Oral                | LD50 Dermal | LC50 Inhalation                           |
|------------------|--------------------------|-------------|-------------------------------------------|
| Paraformaldehyde | LD50 = 800 mg/kg ( Rat ) |             | LC50 = 1070 mg/m <sup>3</sup> ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 1

## (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

Category 1

May cause sensitization by skin contact

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; Category 2

Possible cancer hazard. May cause cancer based on animal data

(g) reproductive toxicity; No data available

(h) STOT-single exposure; Category 3

Results / Target organs

Respiratory system

(i) STOT-repeated exposure; No data available

Target Organs

No information available.

(j) aspiration hazard; Not applicable  
Solid

## Symptoms / effects, both acute and delayed

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting; Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## SECTION 12. ECOLOGICAL INFORMATION

## Ecotoxicity effects

Do not empty into drains. .

| Component        | Freshwater Fish | Water Flea         | Freshwater Algae | Microtox |
|------------------|-----------------|--------------------|------------------|----------|
| Paraformaldehyde | >10 mg/L 96h    | EC50 = 42 mg/L 24h |                  |          |

Persistence and Degradability  
PersistenceExpected to be biodegradable  
May persist, based on information available.

## Bioaccumulative Potential

Bioaccumulation is unlikely

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**Mobility in soil** Is not likely mobile in the environment due its low water solubility

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains.

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

**UN-No** UN2213  
**Proper Shipping Name** PARAFORMALDEHYDE  
**Hazard Class** 4.1  
**Packing Group** III

IMDG/IMO

**UN-No** UN2213  
**Proper Shipping Name** PARAFORMALDEHYDE  
**Hazard Class** 4.1  
**Packing Group** III

IATA

**UN-No** UN2213  
**Proper Shipping Name** PARAFORMALDEHYDE  
**Hazard Class** 4.1  
**Packing Group** III

**Special Precautions for User** No special precautions required

## SECTION 15. REGULATORY INFORMATION

**International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component        | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|------------------|-----------------------------------------------------|-----------------------------------------|------|-------|--------|------|-----|-------|------|------|------|----------|
| Paraformaldehyde | X                                                   | X                                       | X    | X     | -      | X    | X   | X     | X    | X    | X    | KE-27818 |

## National Regulations

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 11-Oct-2010  
**Revision Date** 13-May-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**